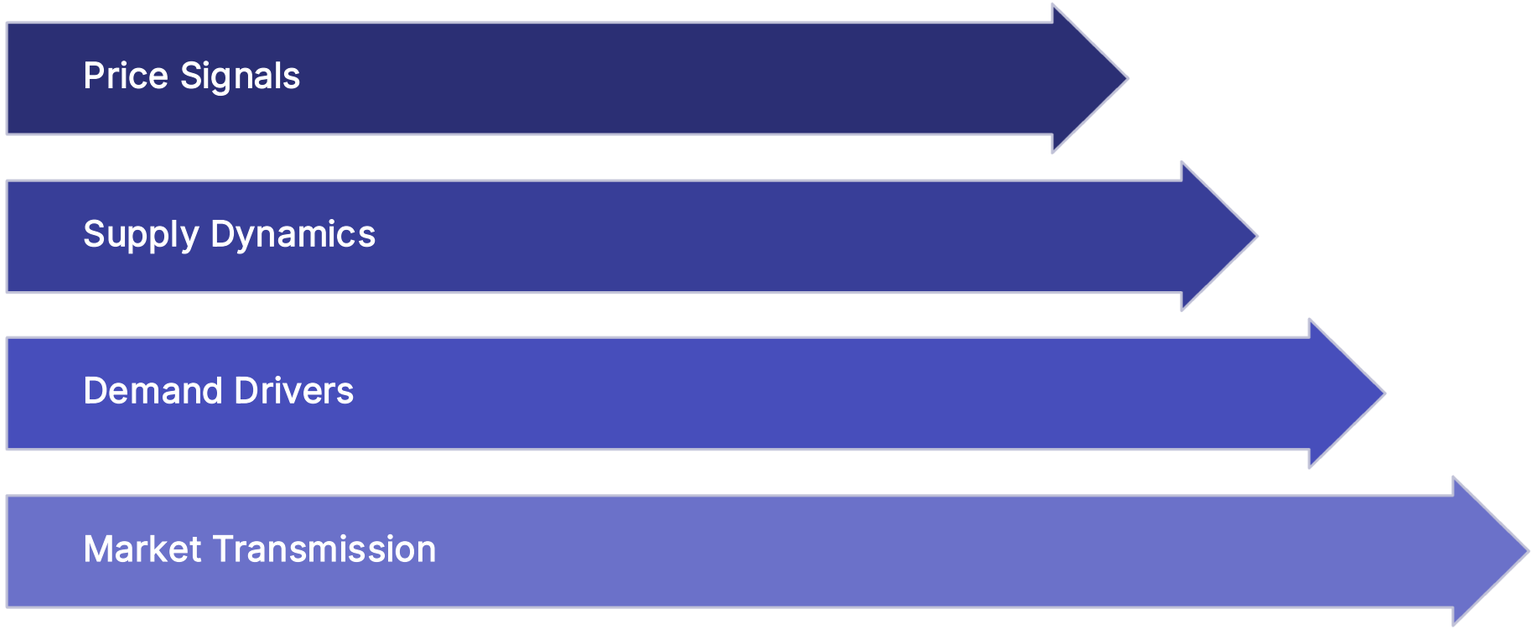
Understanding Global Markets Through Commodity Price Behaviour



Problem Statement

- How have global commodity prices evolved over the last six decades?
- Do commodity prices reflect long-term inflation trends?
- How do different commodity markets behave (oil, gas, metals, food)?
- Are price movements globally synchronized or region-specific?
- What impact do major economic events have on commodity prices?

Dataset Overview

Our analysis of global commodity price behavior is built upon a comprehensive and robust dataset.

Data Source & Scope

- **Source:** World Bank – Commodity Price Data (Pink Sheet)
- **Time Period:** 1960 – 2026 (Monthly Data)
- **Data Coverage:** 70+ commodities across energy, metals, and food categories

Key Data Challenges

- Missing values
- Inconsistent formats across columns
- Required extensive preprocessing before analysis

Methodology and Tools

Our robust methodology ensures the integrity and depth of our analysis, employing industry-standard tools and techniques.

Data Processing

Cleaned and structured raw data using `Python(Pandas)`. We meticulously handled missing values and inconsistent formats to ensure data integrity and reliability for subsequent analysis.

Data Analysis

Leveraged `SQL (PostgreSQL)` for efficient data aggregations and transformations. We applied statistical functions such as `AVG` and `STDDEV` to perform comprehensive trend and volatility analysis.

Visualisation

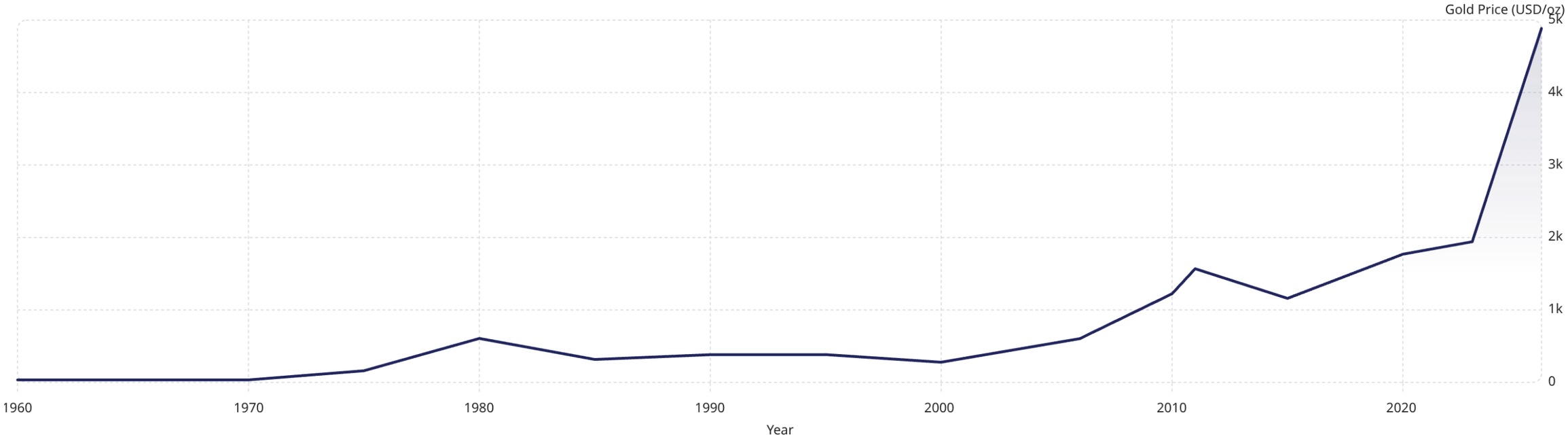
Built interactive dashboards in `Power BI` to present our findings clearly and engagingly. Designed dynamic time-series charts for easy identification and interpretation of price trends over time.

Analytical Approach

Conducted detailed year-wise and comparative analysis across different commodities. This allowed us to link observed price movements with significant real-world economic events, providing valuable contextual insights.

INFLATION GOLD

Gold Price Trends & Inflation



Long-term upward trend indicate a strong inflation signal.

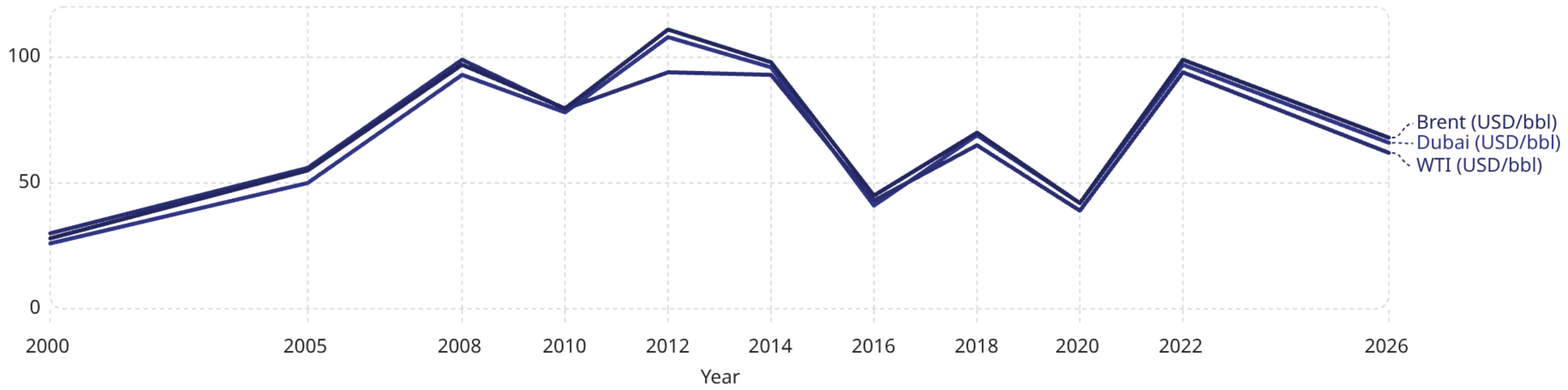
A sharp spike in 1980 reflects market shocks and high inflation of the era.

Rapid growth post-2006, coinciding with global financial instability.

Peaks in recent years signal the current high inflation environment.

Gold acts as a long-term hedge against inflation

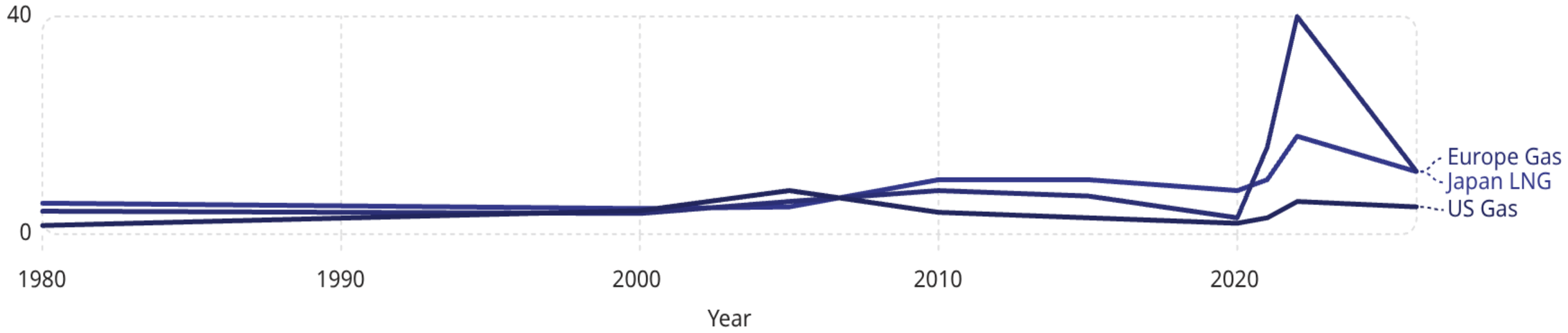
Oil Comparison (Brent vs Dubai vs WTI)



- Brent crude oil typically maintains a higher price point compared to Dubai crude due to its lower sulfur content and ease of refining.
- A significant peak in global oil prices was observed around 2012, reflecting geopolitical tensions and strong demand.
- WTI (West Texas Intermediate) prices show similar general trends and fluctuations to Brent, but often trade at a discount due to transportation costs and quality differences.

Global oil prices exhibit synchronized movements, yet quality differences and regional market dynamics lead to distinct pricing benchmarks.

Natural Gas Markets: Regional Price Behavior

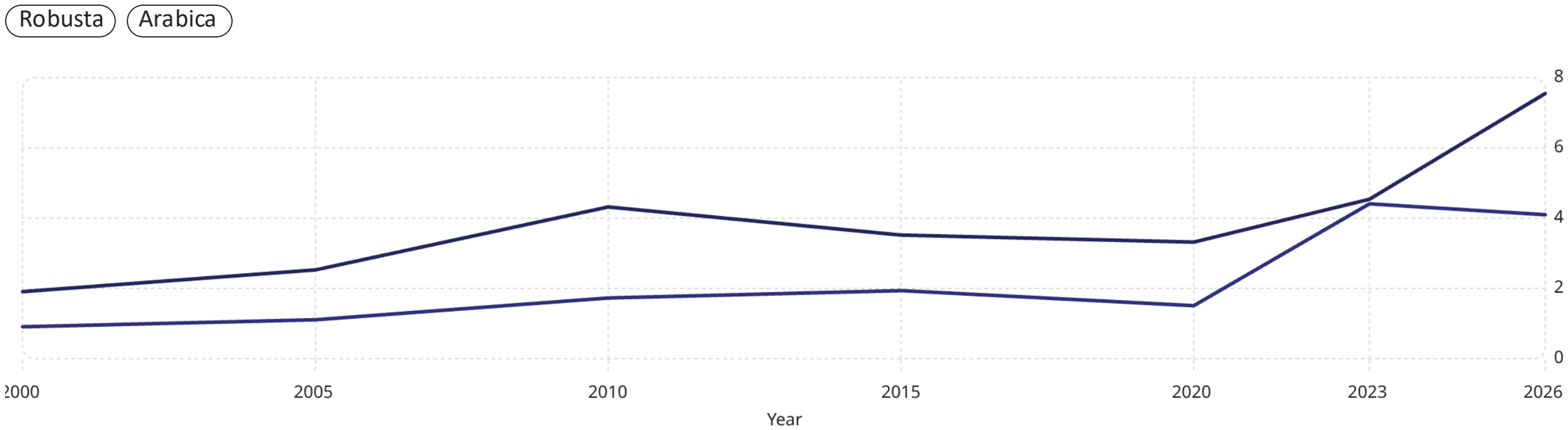


- US natural gas prices remain relatively cheap and stable due to abundant domestic supply.
- Japan's LNG imports consistently command higher prices, reflecting its reliance on global shipping.
- European gas markets experienced extreme volatility, with a major price spike around 2022 driven by geopolitical events.

Natural gas markets are region-specific, unlike globally unified oil markets.

COFFEE

Coffee Analysis



- Both Arabica and Robusta coffee prices have shown a significant upward trend over the past two decades, reflecting growing gbbal demand.
- Arabica, the premium coffee bean, consistently maintains a higher price point compared to Robusta due to its superior flavor profile and cultivation requirements.
- The price gap between Arabica and Robusta has notably increased over time, indicating a faster growth rate for premium coffee varieties.

Premium commodities exhibits faster price appreciation due to heightened consumer demand and increased market sensitivity to supply shocks.

FOOD INFLATION

Global Food Inflation

Understanding the dynamics of food commodity prices is crucial for assessing global economic stability and living standards.

- Food commodity prices show a consistent rising trend over time, reflecting growing global demand and production costs.
- Sharp price spikes are evident during periods of global economic stress, such as the 2008 financial crisis and the COVID-19 pandemic, indicating market sensitivity to external shocks.
- This upward trajectory and volatility intensify pressure on food affordability worldwide, disproportionately affecting vulnerable populations.

Food prices reflect global economic stress and supply chain disruptions.

Conclusion: Interconnected Markets & Inflation

Inflation Trends

Commodity prices consistently reveal long-term inflation patterns.

Market Diversification

Commodity markets exhibit diverse behaviors, ranging from globally synchronized to distinctly regional.

External Impacts

Major external events are powerful drivers of significant price volatility across commodities.

Data reveals that inflation and market dynamics are deeply interconnected across commodities.